

Intro to Software Engineering – Quiz #2

June 10, 2023

Quiz #2
Total: 100 points
Due Date: June 24th, 2023

*This quiz must be completed individually.
I.e. you should work alone without any help or collaboration from others.
Please provide detailed answers to the following exercises.*

Overview

In this lab, you will use Neo4j to analyze a supply chain network. You will create a graph database from scratch, perform various queries to gain insights and visualize the results. The dataset includes information about suppliers, manufacturers, distributors, and products. The goal is to gain insights into the flow of goods, identify key suppliers, and understand the network's structure. Each question outlines the grading scheme, your total grade will be capped at 100 points.

You should submit:

1. A pdf report including:
 - Your name and student number
 - The Cypher queries used and the results obtained.
 - A description of the data model and the rationale for your design choices.
 - Any additional insights or observations made during the analysis.

Dataset

You will create your dataset based on the following entity relationships.

Entities

- **Suppliers:** Name (String), location (String), reliability score (Float - 0.0 to 1.0)
- **Manufacturers:** Name (String), location (String), capacity (Int - Kg)
- **Distributors:** Name (String), location (String), delivery efficiency (Float - 0.0 to 1.0)
- **Products:** Name (String), category (String), production cost (Int)

Relationships

- **SUPPLIES**: From Suppliers to Manufacturers (with attribute: transportation cost - Int)
- **MANUFACTURES**: From Manufacturers to Products (with attribute: production time in hours - Int)
- **DISTRIBUTES**: From Distributors to Products (with attribute: delivery time in hours - Int)

Question 1 Data Ingestion and Graph Creation (20 marks):

Task: Create a Neo4j database and import your supply chain dataset. Define appropriate labels and relationships with attributes.

Instructions:

- Create nodes and relationships as per the descriptions given above.
- Include a brief description of the data model in your submission.

Grading Scheme: 20 points

- Correct labels and relationships with attributes: 10 points
- Successful data creation: 10 points
- Clear data model description: 5 points

Question 2: Identify Key Suppliers Based on Reliability and Cost (20 marks):

Task: Identify the supplier that supplies the most manufacturers, weighted by reliability and transportation cost.

Instructions:

- Consider both the number of manufacturers and the reliability score of suppliers.
- Incorporate transportation costs into the evaluation.

Grading Scheme: 20 points

- Correct query: 15 points
- Correct result: 5 points

Question 3: Find Top Manufacturers by Capacity and Production Time (20 marks):

Task: Find the top 3 manufacturers based on their capacity and average production time of products.

Instructions:

- Calculate the efficiency of manufacturers as capacity divided by average production time.

Grading Scheme: 20 points

- Correct query: 15 points
- Correct result: 5 points

Question 4: Analyze Distribution Efficiency (20 marks):

Task: Determine the distributor that distributes the most products efficiently, considering delivery time and efficiency score.

Instructions:

- Consider both the number of products distributed and the delivery efficiency score.

Grading Scheme: 20 points

- Correct query: 15 points
- Correct result: 5 points

Question 5: Identify the Most Expensive Product to Produce (20 marks):

Task: Identify the product with the highest total production cost, including transportation, production, and delivery costs.

Instructions:

- Calculate the total cost for each product as the sum of transportation cost from supplier to manufacturer, production cost, and delivery cost from manufacturer to distributor.

Grading Scheme: 20 points

- Correct Query: 10 points
- Correct Result: 10 points

Bonus: Supply Chain Optimization (20 marks):

Task: Optimize the supply chain path from the supplier to the distributor for a specific product, considering all attributes.

Instructions:

- Identify the best path from supplier to distributor for a given product that minimizes the total cost (transportation + production + delivery) and maximizes reliability and efficiency.

Grading Scheme: 20 points

- Correct query logic: 15 points
- Correct optimization: 5 points